

Chapter 12

Emergence, Downward Causation, and Interlevel Integrative Explanations



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Abstract In this article, I propose a unified account of systemic emergence, downward causation, and interlevel integrative explanations. First, I argue for a relational-transformational notion of emergence and a structural-relational account of downward causation in terms of both its transformational and conditioning effects. In my view, downward causation can avoid the problems traditionally attributed to it, provided that we are able to reconceptualize the notion of ‘whole’ and that form of causality in a purely relational way. In this regard, I distinguish contextual or whole-to-part causation from downward causation, the latter defined by the existence of *second-order* structural relations. Finally, I argue that while emergence and downward-structural causation imply the in-principle failure of micro-determinism and therefore micro-reduction, they do not rule out the possibility of any type of explanation. On the contrary, they call for the development of interlevel integrative explanations.

Keywords Emergence · Downward causation · Integrative explanations · New mechanism · Relational ontology · Second-order relations

12.1 Introduction (to a Relational Ontological Approach)

According to the dynamic relational perspective that I will follow here, systemic emergence and downward causation must be conceptualized in terms of certain transformative and conditioning relations involving wholes, as *systems of relations*, and their proper parts, as *relata* of such relations.

A relational ontological view, as I conceive it, does not postulate that relations are all there is. Rather, it is an ontology according to which every particular entity (independently of whether it is conceptualized as an object, process, activity, event, etc.) owes its *identity* and *existence* to a relation between its *endogenous* and

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exogenous relations involving other entities, including in the context of higher-level relational systems (see Santos, 2015a: 439–442; 2020: 8693–8597).

In this sense, the basic ontological categories are not relations and objects but relations and relata. Objects are just one kind of relatum. Processes, events, and properties also relate with each other – causally, spatially, temporally, functionally, etc. Even relations can be themselves relata, for they relate and interact with each other in *structures*. As a matter of fact, many relations depend, in terms of their very *existence*, both on specific relata and other relations. For instance, gravitational interaction depends both on the existence of masses and relations of spatial distance. Finally, relations and relata should be seen as standing on the same ontological footing. Indeed, unless abstraction is involved, no relatum exists without being related to something, and no relation exists without being a relation between some relata. In this view, there is no room for absolutely intrinsic properties but only for endogenous and exogenous relational properties.

However, this correlation between relations and relata also includes the systems they form. In fact, *any relation between two or more relata immediately forms a system of which the relata are proper parts*. Therefore, relations, relata, and relational systems always come together as three co-relative ontological categories. From this vantage point, it makes no sense to ascribe an *absolute* ontological priority to relata, relations, or the systems they constitute.

This can also be seen from a temporal perspective. It certainly seems reasonable that there has never been a time in which different individuals existed without being related to each other in some way and without, therefore, being constitutive relata of some relational system. Likewise, there was never a time when systems existed without being themselves structured by the relations between their constitutive parts or relata. For this reason, it also makes no sense to assign an *absolute* temporal precedence to relata, relations, or their systems. From a temporal point of view, reality consists simply of the continuous generation of new relational systems from prior changes in other systems and relations and the ongoing transformation of entities both as relata and as relational systems.

This dynamic relational view thus rejects not only the metaphysical atomism or individualistic essentialism of old mechanistic philosophies, but also the holistic notion of brute or in-principle inexplicable emergent wholes. First, no whole exists apart from (and hence somehow independently of) the complete set of its parts' relations and its relations with the outside world. Second, the complete explanation of every system must always include an account of its formation as the historical outcome of prior transformations in other systems and relations.

I proceed as follows. In Sect. 12.2, I will present a *relational-transformational* account of systemic emergence, which was first elaborated in (Santos, 2015a) and later developed along the lines of a neo-mechanist perspective in (Santos, 2020).

In Sect. 12.3, I will articulate a *structural-relational* account of downward causation by answering the following four questions: what is a whole? (Sect. 12.3.1), what is the 'higher level' of an integrated whole? (Sect. 12.3.2), how should we conceptualize downward causation? (Sect. 12.3.3), and how does downward causation work? (Sect. 12.3.4). In this section, I will distinguish between two types of

causation: contextual or whole-to-part causation and downward causation. I define downward causation in terms of the existence of *second-order* structural relations, which is why I shall call it downward-structural causation.

In Sect. 12.4, I will show that it is the objective existence of systemic emergence and downward-structural causation that ultimately justifies the in-principle failure of micro-determinism and micro-reduction and that, at the same time, demands the use of *interlevel integrative explanations*.

Here lies, in my view, the *positive epistemological significance* of ontological emergence and downward causation. Furthermore, it is also here that we may find the real ontological and epistemological *novelty* of any neo-mechanistic perspective vis-à-vis the *old* mechanistic philosophies.

12.2 Emergence

The notion of systemic emergence has been historically defined in opposition to the notion of lower-level, whole-to-part or micro-reduction. Indeed, from the point of view of part-whole relations, there are only two possible ways of obtaining an absolutely *asymmetrical* or unidirectional relation of reduction: either *macro*-reduction or *micro*-reduction. That is, either we completely reduce the properties and respective relations (including laws) of the proper parts of a given system to the properties and laws *only instantiated* by that system, or we reduce the properties and laws of a system to the *intrinsic* or *system-independent* properties and respective relations (including laws) of its proper parts.

I italicized the phrases ‘only instantiated’ and ‘intrinsic or system-independent’ to highlight the fact that, in order to constitute a purely *asymmetrical* reduction, the reducing term must have *all* the resources required to account for the reduced term *independently* of the latter.

Consider micro-determinism and micro-reductionism. How completely micro-determined and therefore micro-reducible can a system’s property be when some of the parts’ properties contributing to its production are only instantiated *by virtue of* the parts’ integration within that very system? The only alternative form of reduction is a *partial* and *reciprocal* reduction. Yet, since the latter does not constitute an asymmetrical or unidirectional relation, it is compatible with some notions of emergence and downward causation.

My notion of relational-transformational systemic emergence (RTE) is defined in opposition to both complete micro- and macro-determinism and, consequently, to complete micro- and macro-reductive explanations. In my view, all processes of RTE are characterized by the necessary conjunction of two main features: (i) they all involve *relations*, and (ii) they all involve a *transformation* of the *actual* identity of lower-level entities as *parts* of some wholes or as constitutive *relata* of some systems of relations (Santos, 2015a, 2020).

By a change of the *actual* identity of an entity, I mean a change in the set of properties or behaviors that an entity actually has or manifests, even if by gaining

and losing some properties the entity will also change in terms of its *potential* identity, that is, in terms of the powers or capacities associated to such properties.

In this sense, I propose the following characterization of systemic emergence. A property *P* of a system *s* is emergent if, and only if,

- (i) *P* is a property of a specific global organization (*R*) of the proper parts of *s*, and
- (ii) *R*, and hence *P*, are *not* completely determined by (thereby *not* being fully explainable or reducible in terms of) the *intrinsic* or *system-independent* properties, and respective relations and laws, of the proper parts of *s*.

This means that the existence and explanation of property *P* depend on, at least, some *system-dependent relational* properties of the proper parts of system *s* – that is, properties which the lower-level entities only have or manifest *by virtue of* being *parts* of system *s*, or by virtue of being *relata* within the specifically organized system of relations called *s*. It should be noted that the qualifier ‘at least’ was included because *P* may also depend on some relations that the system *s* has with its external environment, including as a proper part of a yet *higher-level* system.

To say that *x* does not completely determine or produce *y* is the same as to say that *x* provides the *necessary* but *not the sufficient* conditions for the ontic determination or production of *y*’s existence or identity (see Bishop & Atmanspacher, 2006; Bishop et al., 2022: 27 and 94; and Santos, 2021:1).

RTE is found in that class of mereological complexes called ‘integrated systems’ (Bechtel & Richardson, 2010: 27), characteristically defined as being only ‘minimally decomposable’, i.e., whereby the ascription of independent behaviors or functions to their proper parts taken separately is impossible (Bechtel & Richardson, 2010: 26–31).

This notion of RTE is suggested by Wimsatt’s claim that emergence must “involve some kind of *organizational interdependence* of diverse parts” (Wimsatt, 1997: S375 – italics inserted; also: 2006: 673). In this sense, emergence implies not only a “dependence of a system property on the arrangements of the parts”, but also, and *above all*, “the context-sensitivity of relational parts’ properties to intra-systemic conditions” (Wimsatt, 2000: 270). In fact, it is important to distinguish between these two conditions and the different notions of emergence they imply. To say that an “emergent property is – roughly – a system property which is dependent upon the mode of organization of the system’s parts” (Wimsatt, 1997: S373), or that the “emergence of a system property relative to the properties of the parts of that system indicates its dependence on their mode of organization” (Wimsatt, 2006: 673), is not in itself sufficient to prove the inadequacy of a micro-reduction.

Emergent properties are not just organizational, collective, or non-distributive systemic properties. Indeed, metaphysical atomism has always recognized that many systems’ properties are dependent on specific modes of organization, combination, or arrangement of their parts. The issue is that for metaphysical atomists, any organizational property of a system is completely reducible to the *intrinsic* properties of its parts and their respective laws and causal relations. In this precise sense, organization is not enough. An organizational property can only be taken as a real emergent property if the *organization* is not itself completely determined by

the *intrinsic* properties and respective laws of the lower-level entities composing that organization (Santos, 2015a: 431–439; and Santos, 2020: 8690–8693).¹

The fundamental difference between RTE and mere ‘organizational emergence’ is that in the latter, intra-systemic relations and global organizations only intervene in the construction of the existence and identity of the *systems* taken as wholes, while in the case of RTE they also intervene in the construction of the identity, if not the existence itself, of the systems’ *parts*.

Some wholes may thus be said to be different from the mere sum of their parts, not only in the sense that they also depend on a specific organization of the parts, but, as Scott Gilbert put it, “in the sense that *the properties of each part* are dependent on the context of that part within the whole in which it operates” (2010: 618 – italics inserted).

In the important new introduction to the 2010 second edition of their book *Discovering Complexity*, Bechtel and Richardson clearly suggest the notion of RTE when identifying two basic conditions for a mechanistic notion of emergence that is “*neither weak nor epistemic*” (2010: xlv – italics inserted; see Santos, 2020: 8700–8701).

First, the activities or operations of the system’s parts depend on the actual behavior and the causal capacities of the other parts in a cyclic (non-sequential) and nonlinear way, and “to the extent that feedback is systemwide, these dependencies will result in operations that are specific to the system”. This condition refers to the fact that “the behavior of the components is system dependent” (2010: xlv). Secondly, “the nonlinearities affecting component operations must in turn affect the behavior of the system” (2010: xlv). As Bechtel and Richardson note, when these two conditions are met, “the systemic behavior is reasonably counted as emergent, even though it is fully explicable mechanistically” (2010: xlv–xlvii).

In this regard, the idea that ontic emergence is “spooky” (Craver & Tabery, 2019) or “suspect” because it suggests a “discontinuity” between a system and its “parts, activities, and organizational features of the system in the relevant conditions” (Povich & Craver, 2018: 190) conflates features that should be distinguished so as not to render meaningless the very contrast between reductionist and interlevel integrative explanations. In fact, the ‘spooky’ or ‘suspect’ character of ontic emergence is just a consequence of ascribing an *absolute* meaning to the notion of ‘discontinuity’ between a system and its parts. Again, the question is whether parts considered in isolation, i.e., as *independent* individuals with all their alleged absolutely *intrinsic* properties, provide *both* the *necessary* and *sufficient* conditions for the

¹For an excellent depiction of atomism, both in its ontological and methodological aspects, see Humphreys (2016: 1–26). The notion of transformational emergence that I proposed in (Santos, 2015a) differs from the transformational perspective elaborated by Paul Humphreys (2016) in two essential ways. First, my notion of emergence applies to part-whole relations, thereby being explicitly based on a distinction between different levels of organization. Second, in my view, all transformation processes are caused by and must then be necessarily explained in terms of specific *relations*. This is the reason why I have been arguing for a *relational*-transformational notion of *systemic* emergence.

ontological determination of all properties and behaviors that they may exhibit in all possible relational contexts or systems, as well as for the ontological determination of all systems of which they can become parts.

Assuming the *realist* view that “the direction of explanation recapitulates the direction of determination” (Klee, 1984: 60), ontic systemic emergence simply means that we do not find both the necessary and sufficient conditions for the ontological determination and thus for a complete explanation of a given property of a system at the level of the *intrinsic* or *system-independent* properties of the parts of that system, and their respective relations.

RTE can occur, of course, either during the development of a system or in the generation of new systems. Furthermore, any system, just like any of its parts, also acquires properties by virtue of its exogenous relations with other systems, including as a *part* of or *relatum* within a further *higher-level* system of relations. As a matter of fact, it is the *existence* of such interlevel relations of determination that justifies the need for the explanatory task of ‘situating’ mechanisms or systems in their environments (Bechtel & Richardson, 2010). As Bechtel has stressed, the explanation of any mechanism always requires the consideration of “its relation to conditions in its environment” (Bechtel, 2011: 538), “including its incorporation within systems at yet higher levels of organization” (Bechtel, 2006: 40–41).

As it was said, RTE is defined in opposition to both complete micro- and macro-determinism and, consequently, to complete macro- and micro-reductive explanations. But to deny such forms of determination and explanation is *not* the same as to deny any form of determination and explanation. If the right relations are identified and the transformations they cause are taken into account, we can regard any emergent property as completely determined, and its complete explanation can be, at least in principle, provided. In Sect. 12.4, I will specifically address this issue when dealing with interlevel integrative explanations.

In the next section, I will show how RTE relates to downward causation, thereby clarifying the way in which parts can be determined by their wholes with the help of some concrete examples.

12.3 Downward Causation

Following the relational viewpoint presented here, I claim that there is only downward determination (causal or otherwise) if the *relations* which determine the parts of a given whole are at a genuinely *higher level* than those parts.

I construe downward causation (DC) as a particular form of downward determination because systems may determine their proper parts by means of both *causal* and *non-causal* relations. For example, parts may acquire or lose some causal powers without undergoing any causally induced inner structural changes if such powers are to be considered genuine extrinsic relational properties; parts’ behaviors may be constrained by the topology of their systems’ structures, and so on.

The parts of a system may alter their *actual* identities or only their *potential* identities as a result of downward determination. By a change of the actual identity of an entity, I mean a change in the set of properties or behaviors that it actually has or manifests, even if, by gaining and losing some properties, the entity will also change in terms of the powers or capacities associated with such properties. By a change of the potential identity of an entity, I mean a change in its set of powers or capacities (without considering their actual manifestation) or in terms of a reduction or extension of its degrees of freedom.

The downward causal determination of the actual identity of the proper parts of a system may be called *downward causal transformation*. The downward causal determination of their potential identity may be called *downward causal conditioning*. Furthermore, since parts may be changed by acquiring or losing powers, they may be subject to downward causal conditioning, either in *empowering* or *constraining* terms (Archer, 1995; Hooker, 2013: 761).

Finally, because DC is a causal relation, it must necessarily be seen as a *diachronic* process. This means that even downward causal conditioning is never, strictly speaking, synchronic. To say that part *x* of system *s* is conditioned by *s* at time *t* is to say that *x* can only act this way or that way at any time *after t*, thereby changing the set of its possible future behaviors. For example, to say that part *x* has lost a given power or capacity *P* at time *t* is to say that *x* cannot act in a *P*-way *from that moment on*, i.e., after *t*. At any time *t*, every entity is *just acting* given the possibilities defined *before t*. The effects of causal conditionings thus always come *after* the imposition of such conditions.

The notion of DC has two well-known problems. The first problem is that it contradicts the principle of causal closure or completeness of the micro-physical level of reality and its associated principle of overdetermination. The second problem is that it seems to contradict the notion that causal relations must be non-reflexive. I will address these issues by answering the following four questions: what is a whole? (Sect. 12.3.1), what is the ‘higher level’ of an integrated whole? (Sect. 12.3.2), how should we conceptualize DC? (Sect. 12.3.3.) and how does DC work? (Sect. 12.3.4). In the following, I propose a structural-relational account of DC, which can avoid the problems traditionally attributed to it as well as allow it to be easily placed within a neo-mechanistic framework.

12.3.1 What Is a Whole?

The main reason for the troubles that the notion of DC faces when confronted with the classical notion of causal relations as *non-reflexive* lies, in my view, in the very notion of ‘whole’ that has been (more or less tacitly) adopted.

If a whole, in its broader sense, is just a *set* of parts and their respective relations, how can that *set* causally affect its *subset* of parts? According to the received view, causal relata must be spatially and temporally distinct, and thus not related compositionally. Part-whole relations, in turn, should be limited to compositional

relationships, meaning those of ‘constituting’ and ‘being constituted by’, either in purely spatial or mechanistic terms (working parts, component operations, etc.). Thus, how can a whole causally influence or condition its constituent parts?

In my view, part-whole causal relations should not be seen as relations in which a whole, taken as a *set* of parts and their respective relations, causally interacts with, thereby affecting, the *subset* of its parts. Their causal relations should neither be thought of as relations between two object-like things: the whole as an object *vs.* its parts as a plurality of objects. Both views are conceptually flawed, thereby creating unnecessary problems.

If by the term ‘whole’ (or system) we simply mean a set of elements as proper parts and their respective relations, then any whole has two correlative but distinct dimensions: the set of its proper parts and the set of its proper relations. Therefore, the proper parts of a whole are just the *relata* of the *relations* constituting that whole. In this view, *any relation between two or more relata immediately forms a whole of which the relata are proper parts*. In short, a whole is to its *proper parts* what a system or network of relations is to its constitutive *relata*.

Of course, there are different kinds of wholes, depending on the nature of their constitutive relations, the degree (if any) of their organization and interdependence, their stability or persistence conditions (some are highly transient, while others are very stable), etc. But the above characterization stands as the most general notion of a whole in relational terms.

How can then wholes causally interact with their parts? To address this question, I find it helpful to distinguish between two different *perspectives* of a whole. I shall call them the *outside* and the *inside* view. From the outside view of a whole, a whole is just a set of entities, as its proper parts, and their relations. From the *inside* view of a whole, we should say that for every part taken as a *relatum*, *its* whole is just the set of relations among *all its co-relata*. In this sense, a whole is nothing more than the ‘relational context’ in which an individual is embedded.

To ask, then, whether a part is affected by its whole is just to ask whether an individual entity is affected *by being a relatum* in a given *system* or *network of relations involving all its co-relata* (Santos, 2015b). This is to say that for each part, there is a different whole taken as a system of relations among all other co-relata. Only the outside view ‘presents’ a unique whole.

We can thus make sense of whole-to-part causation without invoking the holistic notion of a whole as an unanalyzable individual or primitive thing. To be part of a whole is simply to be a *relatum* within a specific system of relations. Thus conceptualized, whole-to-part causation is no longer a reflexive relation.

As a matter of fact, this relational perspective has already been advocated by Jean Piaget, in 1950, while addressing the relationships between sociological and psychological explanations (a problem classically polarized by the holistic and individualistic views). In the context of that analysis, Piaget poses the following question: “If the individual is the element and the society is the whole, how is it possible to conceive a totality which modifies the elements which make it up, without making use of other material than these elements themselves?” (1995: 39). Piaget’s answer was that the notion of social totality, or whole, should not be conceived as

“a combination of pre-existing elements”, nor as a “novel entity” (in the sense of something existing over and above its parts), but as “*a system of relationships*, each of which in its own right brings about a *transformation* of the elements thus related” (1995: 41 – italics inserted).

It is only because it is often assumed (even without full awareness) a holistic, mystifying notion of whole, that whole-to-part causation is frequently seen as a highly unique type of relation endowed with a certain air of mystery. As soon as we demystify the notion of whole, we can easily see how widespread whole-to-part or contextual causation is in nature.

12.3.2 What Is the ‘Higher Level’ of an Integrated Whole?

Let us then assume that it is not the whole as an individual thing or *object* that can causally affects its parts taken as a collection of further objects. Instead, the *system of relations* constituting a whole causally affects each of its constitutive *relata* as proper parts of that system. But in a system of relations, what stands at a *higher* level than the *relata* constituting that system? In my view, the higher level of any non-aggregative system is simply the level of the *global organization* of all its proper parts, as well as the properties and laws of that organization taken in and of itself.

A distinction must nevertheless be made between ‘component systems’ and ‘integrated systems’ (Bechtel & Richardson, 2010: 26–27). The organizing relational structure of a component system is constituted by merely quantitative and combinatorial relations between quasi-independent parts. On the contrary, the organizing relational structure of any integrated system is constituted by some qualitatively *transformative* and *interdependent* relations between its proper parts (see Santos, 2020: 8697–8700).

The fact that the relevant inter-individual relations within a given system are *interdependent* means that they do not occur or develop independently of each other, which means that they are not conceivable as separate or atom-like dyadic relationships. Relations do not come one by one, acting separately from the others, and affecting one-after-another each relatum at a time. Integrated systems are systems of interdependent parts, which means that their behaviors and relations are also interdependent. This is why integrated systems’ parts are said to be only ‘minimally decomposable’, as it is impossible to ascribe ‘independent’ properties or causal works to them (Bechtel & Richardson, 2010: 27, 31).

But it is not only that the inter-individual relations between the parts of an integrated system are dependent on each other. The key feature is that their interdependence follows specific *system-level modes of organization*. That is, parts’ relations are not dependent on each other in a purely haphazard, contingent, or arbitrary way. For example, in eukaryotic cells, protein folding always takes place *after* translation; transcription always takes place *before* translation; mature mRNAs are always translated *outside* of the nucleus where this process always involves the causal

intervention of ribosomes in *another region* of the cell called cytoplasm. All these inter-individual relations are not dependent on each other solely in terms of their *doings* and *outcomes*; they also follow a *system-specific global order or organization*. In sum, the higher level of any integrated system is made up of two different kinds of *second-order relations*, namely, relations of *systemic interdependence* and relations of *lawful interdependence*.

Now, these specifically organized systems of relations are at a clearly *higher level* of organization than the *inner* organization of *each* of their constitutive relata as lower-level subsystems. Furthermore, the mode of organization of any integrated system – which always involve specific re-equilibration or self-regulation causal processes (homeodynamic and allodynamic), as well as topological and temporal relations – is always *new* and *different* from the inner modes of organization of its parts.

12.3.3 *How Should We Conceptualize Downward Causation?*

From what has been argued, it follows that DC must be conceptualized as referring to the set of transformations and conditionings that a *specifically organized system of interdependent relations* exerts in each of their relata as lower-level parts of that system.

Therefore, DC implies the existence of *second-order relations* that structure or organize in a *specific way* (causally, spatially, and temporally) the *first-order* relations between the parts of a system, thereby defining the *way* in which these first-order relations determine and change the parts. In other words, DC does not apply to cases where an individual is just causally affected by a set of first-order relations with other individuals but to cases where individuals are causally affected by the *way* their first-order, inter-individual relations are *themselves related* (structurally and functionally) in a *systematic* manner.

This only reinforces the need to distinguish the *macro-relational structure* of a system from its *micro-compositional structure*, i.e., from the set of all properties and first-order, inter-individual relations among the system's parts (Santos, 2020: 8697–8698).²

Some first-order relations can only exist if organized in a particular way (can one imagine translation occurring before, rather than after, transcription in any cell?).

²As Peter Simons has written, “[a structure is] the *total or overall* relation of all the parts, as distinct from the multitude of binary and other relations between and among the parts. In fact, we need even to consider the various kinds and properties of the individual parts as part of this overall relation or pattern. In algebra or model theory, it is usual to define a ‘relational structure’ as a *sequence* $\langle D, R_1, R_2, \dots \rangle$ where D is the domain (set of parts) and R_1, R_2 , etc. the specific relations among these elements. The structure is here as it were a sort of *super-relation* on the domain and relations” (1987: 355–356). For an in-depth and comprehensive understanding of the notion of structure and its key role in the formation of contemporary science, see Piaget, 1971b.

In some systems, certain relations can be organized in different ways (within certain limits, of course), but their causal effects will also be different. In any case, what *causally affects* (transforms and conditions) each part of an integrated system is not a sum of independent inter-individual relations but a *specifically organized set* of them. To put it differently, in integrated systems, the *causal workings* of inter-individual relations cannot be separated (except through abstraction) from *how* they are specifically structured or organized. The specific modes of organization (i.e., relational structures) of integrated systems must thus be counted as real contributing *causes* of the behavior of those systems' parts. Inter-individual causal relations and their distinctive modes of organization via some second or higher-order relations do not operate separately; they *come together, work together and act together*.

It should go without saying that the concept of causation associated with second-order structural relations cannot be understood in terms of the concept of *efficient* causation for the very obvious reason that the latter was designed to deal exclusively with first-order, inter-individual relations. While all inter-individual causal relations act as *efficient* causes, their specific modes of organization and interdependence act as *downward structural* causes (see Lawson, 2013: 287; and 2019: 38, 87–88, 199–200, 214–219).

A further distinction may be worth emphasizing. While in contextual or whole-to-part causation, the whole that causally acts on each of its parts can be said to constitute a mere *plurality* (viz., the set of *all other parts'* relations; see above, Sect. 12.3.1), in the case of downward-structural causation, the whole constitutes a genuinely new *individual* due to the structural and functional *unity* (or interdependence) between its constitutive relations. As Simons has noted, it is important “to distinguish between a collection of many individuals and the one individual they compose, if they do” (2006: 599, n. 4). However, as Tony Lawson rightly pointed out, it is the ‘organizing relational structure’ of a composite whole, rather than the whole-as-a-whole, that may be said to exert top-down or downward causation (2013: 287; 2016: 431–432; 2019: 38, 74 n. 9, 214–219).

From this perspective, it is possible to discern the occurrence of DC in any integrated system, whether hierarchically or heterarchically organized. Indeed, even in the most strongly hierarchical systems, the ultimate ‘master controller’ (so to speak) of the parts’ behaviors is always the specifically organized set of relations that structure those systems. In the last instance, it is never a specific part that downwardly causes the other parts. The real agents of DC are always specifically organized systems of relations because it is ultimately *by virtue of* them that some part may have a more relevant causal role in determining or regulating the activities of the other parts.³

The same goes for systems where power or control is more equitably distributed. Even though the powers that each part has are obviously powers *of* each of these

³For a distinction between different notions and uses of the concept of ‘hierarchy’ in systems and network neuroscience, see Hilgetag and Goulas (2020). As the authors show, “diverse ‘hierarchical’ concepts lead to different interpretations of the empirical data, with diverging functional implications” (2020: 8).

parts, the ultimate *source* of their instantiation is always a function of the interplay between each part's inner structure and the organized set of relations that this part has with all the other parts as its co-relata. Indeed, only in a fictional world of abstract individuals would entities have or lack powers exclusively in terms of their inner structures.

12.3.4 *How Does Downward Causation Work?*

I argued that DC should be conceptualized as the set of transformations and conditionings that a specifically organized system of interdependent causal relations exerts on each of their relata as lower-level parts of that system. Furthermore, as I noted above, because DC is a type of causal relation, it must necessarily be viewed as a *diachronic* process. At any given point in time, the organizing relational structure of a system affects or partially determines the behaviors that the parts of that system will or may instantiate at a later time. Likewise, at any time *t*, every entity is just acting given the *possibilities* defined *before t*.

The same is true of upward causation. The individual behaviors of each system's part *causally* contribute to the maintenance or modification of the collective behaviors of the system (Santos, 2015b), thereby contributing to the 'reproduction' or 'transformation' of the system's structure (Archer, 1995; Lawson, 2019). But at any given point in time, the behaviors of the parts simply "collectively constitute (along with the relevant organising structures)" the behaviors of their system (Lawson, 2019: 217).

Therefore, parts do not change because the wholes they compose change, as a mere mereological consequence of being parts of such wholes (Craver & Bechtel, 2007). Parts change because they are relata within specifically organized systems of interdependent transformative and conditioning relations.

This is well exemplified in self-organization processes. Properly speaking, self-organization is a process by which a system *reorganizes* itself in terms of the relations between its parts as a result of some external disturbances that threatened the original organization. If the reorganization process succeeds in 'assimilating' such disturbances, the system will then persist (Atlan, 1979: 165–170; 2011a, b). Rayleigh-Bénard convection, but also immune systems, are two of the most well-known and studied examples of such dynamics (Atlan & Cohen, 2006; Atlan, 2011a, b; Bishop, 2008; Bishop et al., 2022: 37–43; Cohen et al., 2016). The new systemic properties generated by self-organizing processes are thus emergent in a relational-transformational sense (see above, Sect. 12.2).

Most of the behaviors that entities exhibit as lower-level parts of integrated systems can only be explained by the fact that they are parts of such systems, thereby being determined by the structural organization of their constitutive relations. In any organized system of interdependent causal relations, the effects are propagated or transmitted in a specifically ordered manner, with each part thus being both *directly* and *indirectly* related to all other parts' relations in a system-wide way.

For example, although it takes DNA and RNA to produce proteins, it takes proteins (e.g., transcription factors) to regulate the activity of DNA and RNA. Proteins, in turn, will play a key role in manufacturing (e.g., RNA polymerases) the very nucleotide sequences that code for specific sequences of amino acids from which new proteins will then be produced. When two integrated system's parts interact, they are of course the direct causal agents of their own interaction. Yet that relation is *directly* and *indirectly* related to the globally organized set of relations involving the other parts of that system. For example, in eukaryotic cells, the interactive process of translation directly involving mRNAs, tRNAs, and ribosomes is *indirectly* but necessarily related to the outcome of the transcription relation, which directly involves DNAs and RNA polymerases, as well as all other subsystems which contribute to the editing and correction of transcription errors (Vecchi, 2020a).

Consider the well-known causal contribution of chaperons to the protein folding process. Polypeptides and chaperons are, of course, the direct causal agents or interactants of their own relation, but that relation itself depends, both *directly* and *indirectly*, on many other cell-specific types of relations. It is, of course, understandable that when highlighting the importance of the specific causal contribution of chaperons to the process of protein folding, we limit our analysis to *their* causal interaction. When focusing on, and thus conceptually abstracting, that *pair* of interactants, it seems that it is *all about them*. But that can only be done at the cost of a necessary, but highly selective, abstraction, leaving outside many other intra-cellular causal factors (e.g., water molecules, prosthetic groups, osmolytes), without which the relation between chaperons and polypeptides would *not take place* (Santos et al., 2020) – as a matter of fact, without which those proteins would *not even have come into existence*.

This kind of dynamic illustrates the *highest* degree of individuals' dependence on specifically organized systems of relations, that is, a dependence not only in terms of their behaviors or identity but in terms of their own *existence*. No individual comes into existence without having been generated by some system of relations. No individual can persist independently of the interplay between its endogenous and exogenous relations with a specific environment. And many individuals cannot even exist and persist except as parts of specific systems. This is what happens in the particular subclass of integrated systems that Richard Levins has called 'evolved systems', that is, systems "in which the component subsystems have evolved together" (1970: 76).

For example, ribosomes and mitochondria cannot persist as functional structures outside of cells. DNA molecules can but at the cost of becoming just one of "the most nonreactive, chemically inert molecules in the living world" (Lewontin & Levins, 2007: 239). To say, then, that some polypeptides only acquire their native structure by virtue of their relations with other proteins, such as chaperons, means that some *products* of the cell system (i.e., polypeptide chains) only acquire their native structure by virtue of some causal interactions with other *products* of the cell system, called chaperons. Both *relata* are what they are and *act* and *interact* the way they do by virtue of being *constructs* and *relata* of specifically organized systems of relations called eukaryotic cells, involving other *relata*, such as proteins, DNA, and RNA molecules.

This means that the *inner* organizational structures of the lower-level entities provide only the necessary but not the sufficient conditions for the ontological determination of the properties, behaviors, and causal powers that they have and actually manifest in the context of different systems of relations. And this also means that, even though inter-individual relations may be the only *empirically observable* relations, we cannot stop the explanation at the level of such relations when we know that these do not come into existence and take place as separate things but are systemically interdependent in a specifically organized manner.

Unsurprisingly, this is a much-debated topic in contemporary theories of sociological explanation. The idea of stopping the explanation of an integrated system, or part of it, at the level of its empirically observable inter-individual relations corresponds to a “flat” ontological view, where “networks remain linkages between nodes instead of networks of relations” (Donati & Archer, 2015: 22), “despite there being no such thing as context-less action” (2015: i). In the light of this new *atomism of relations*, social explanations should only involve “interpersonal relations (the Individualist concept of ‘social structure’)”, as “the social context should be reduced to refer to nothing but ‘other people’” (Archer, 1995: 36, 34).

As Auyang observes, in “advocating the reductive elimination of social concepts, [methodological individualism] mistakes situated individuals for bare individuals and overlooks the causal feedback that society has on individuals”. In particular, it “forgets that citizens are not Hobbesian men mushrooming from the earth; even in their most self-centered mode, they have *internalized* much social relation and conditioning, so that social concepts *have been built into* their characterization” (Auyang, 1999: 121 – italics inserted).

This is also the reason why most if not all properties, actual behaviors, or interactions of integrated systems’ parts cannot be explained in terms of absolutely *intrinsic* potentialities or dispositions.

A typical example is DNA’s property of ‘being a unit of inheritance’ or the property of ‘being a gene’, when conceived as the causal power of a genetic sequence ‘to code for a particular chain of amino acids’ and ‘to contribute to the construction of functional phenotypic traits’ (Santos, 2020: 8703–8705). These causal powers are system-dependent relational properties that DNA molecules and nucleotide sequences acquire only by virtue of interacting with some other relata, such as RNAs and proteins – including the existence of quality control mechanisms that successfully edit transcription errors – in the context of a *specifically organized* set of transformative and conditioning first-order relations, such as transcription, splicing, translation, and protein folding (Strohman, 1997; Shapiro, 2009; Vecchi, 2020a). This is a clear example of the *empowering* effects that downward causal conditioning can have (see above, Sect. 12.3).

Furthermore, the very definition of what a gene is “depends on the properties of the cell in which the DNA is embedded”, since the properties of a cell “are at least partly determined by transcription of DNA, but, in turn, cellular properties also determine which sequences are to be transcribed, in which combinations, and in what order” (Keller, 2010: 30; see also Atlan & Koppel, 1990). As Keller has stressed, “the necessary dependency of genes on their cellular context, not simply as

nutrient but as *embodying causal agency*, is all too easily forgotten” (Keller, 2001: 309 – italics inserted). This leads to the notion that the findings of developmental biology “point neither to cytoplasmic nor to nuclear determination but rather to a complex but highly coordinated system of regulatory dynamics that operate simultaneously at all levels: at the level of transcription activation, of translation, of protein activation, and of intercellular communication – in the nucleus, in the cytoplasm, indeed in the organism as a whole” (Keller, 1995: 29–30).

However, it is still not enough to consider the overall inner structure of an organism, for “the present environment and its history, at the scales of the cell, the person, the group and the biosphere, interact with the genome to determine its expressions and effects” (Cohen et al., 2016: 6). Moreover, “the contribution of a gene to a phenotype cannot always be separated from the contribution of the environment, despite sophisticated calculations, because the interactions between genome and environment are not linear and not additive” (*Idem*). The whole formation of the structural and functional identity of any eukaryotic cell constitutes a prototypical example of a process of downward-structural causation with the intervention of multiple system-wide feedback loops involving both intra- and extra-cellular interactions (Santos, 2020: 8703–8705).

Another example is provided by the fact that the macrostructure and function of proteins in their native or post-folded structure cannot be accounted for solely in terms of the system-independent properties or potentialities of the components of the primary structure as if they were essentially immutable entities (Santos et al., 2020). Polypeptides only acquire some of their potentialities and functions by undergoing a series of *structural transformations* (i.e., the acquisition of their so-called primary, secondary, tertiary, and quaternary structures) through the developmental process of folding by interacting with specific environmental inputs (e.g., pH and temperature) and contingently present substrates (e.g., water molecules and prosthetic groups) in specific cellular and organismal systems. Higher-level cellular, organismal, and environmental systems of relations do not thus merely trigger the manifestation of some potentialities already given *ab initio* but actually play a causal role in the very generation of new powers or capacities (2020: 377–380). This is a clear case of downward causal transformation and conditioning. And this is the reason why there are good reasons to support a relational-construction-based view of protein development and potentialities formation, which in turn requires the analysis of the dynamical interplay between lower- and higher-level organized systems of relations (2020: 363).

The notion of ‘developmental potential’ might be used as another illustrative example. Even though organisms are the units of development, the potential for that development does neither lie entirely in themselves nor in a specific part of them (such as their genomes). The extra-organismal environment must be counted as one of the three necessary, partial, and complementary causal bases for development potential (Vecchi & Santos, 2023). Therefore, if the genome, the developing organism, and the extra-organismal environmental materials are to be counted as proper structures of the causal basis for an organism’s developmental potential, the latter is not a given. Rather, it is the result of an interaction-based construction, a process

sometimes generating genuine developmental novelties. Hence, what would seem, and is indeed often assumed to be, an intrinsic potential or disposition, is in fact a multi-causal-based extrinsic relational potential of organisms *constructed* in the course of their own development (2023: 26).

This is the reason why we ought to endorse a dynamic-constructivist view of developmental potential, as phenotypes are often constructed out of biotic and abiotic environmental materials. As West-Eberhard notes, “due to changes in both genomic and environmental inputs” (2003: 13), as well as because many of the structural and functional changes undergone by the developing organisms are caused by the assimilation, functional integration, and deployment of environmental resources (a process which West-Eberhard calls ‘developmental entrenchment’), “developmental potentialities” themselves “change” during ontogeny (2003: 13 and 500 ff.).

In the following section, I will elucidate how systemic emergence and part-whole relations of reciprocal and partial co-determination necessitate the use of interlevel integrative explanations in a way consistent with a neo-mechanistic approach.

12.4 Interlevel Integrative Explanations

12.4.1 *The Birth of a ‘New Mechanism’ and Its Integrative Explanation Models*

The birth of a neo-mechanistic view in the twentieth century was essentially due to the impact that cybernetics had – particularly on the biological sciences – from the 1940s onward. The real *novelty* of this neo-mechanistic view relative to the *old* mechanistic philosophies can be primarily found in the recognition of a new, systemic form of causality (typically involving cyclic, feedback, feed-forward, and non-additive relations), and in the subsequent overcoming of the most simplistic notions of reduction in scientific explanation.

In this sense, it is easy to understand why the advent of a new mechanistic approach was seen as very good news for all those looking for a naturalistic way to overcome both the *neo*-vitalist and *old* mechanistic views in biological theory.

Norbert Wiener was explicit in recognizing the birth of a new mechanistic view in his 1948 book *Cybernetics*. According to Wiener, the creation of the modern automata represented both the “complete defeat” of Vitalism (indeed, “the whole mechanist-vitalist controversy has been relegated to the limbo of badly posed questions”) and the birth of a new, non-Newtonian mechanistic view in biology. Yet, this “new mechanics is fully as mechanistic as the old”, since “the essential mode of functioning of the living organism” is basically “the same” as that of the modern automaton (Wiener, 2019: 62–63, also: 54).

As Piaget observed in his 1967 *Biology and Knowledge*,

just at the time when biology was freeing itself from its restricting mechanistic ideas, and when some thinkers, confronted with this deficiency in traditional physical causality, were toying with the idea of a return to vitalism and finality, *a complete reelaboration of the mechanistic approach* opened up new perspectives along lines which corresponded exactly to those notions of circular or feedback systems or of cyclic rather than linear causality (Piaget, 1971a: 130–131 – italics inserted).

Regarding this neo-mechanistic approach (new in relation to “the mechanistic approach of old-fashioned physics”), Piaget highlights the importance of Cannon’s notion of homeostasis and, in general, the “rethinking of causality along the lines since followed by cybernetics”, which, in turn, allowed the scientific study of “auto-regulatory” systems, and “an extension of the general idea of organization, seen as a system of transformations” (1971a: 129–131).

In his 1972s paper, “Noise as a principle of self-organization”, Henri Atlan also acknowledged that the onset of cybernetics in the late 1940s prompted the birth of a “new mechanism” that “progressively imposed itself on biology” (2011b: 95–96). Atlan emphasized the discovery of many neo-mechanistic properties, such as the “redundancy of components, redundancy of functions, complexity of components, delocalization of functions”, “adaptability” and “self-organization” (2011b: 96–98).

Now, with this *broadening* of the scientific concept of causality came a *rehabilitation* of causal models of explanation as well, which represented an overcoming of the empiricist prejudices of both positivism and neo-positivism (see Bunge, 1959). Some of the neo-mechanistic views that would be developed throughout the 1950s, 1960s and 1970s are, of course, in some crucial respects different from the views which would be promoted, from the 1990s onwards, by the so-called “Chicago Mechanists” (Wimsatt, 2018). For example, according to both Piaget and Bunge, although causal explanations are a necessary step of scientific research in the non-formal sciences, they necessarily *depend* on the *previous* discovery and coordination of laws as statements of general facts or repeatable relations. Accordingly, causal *explanations* – unlike the *search* for the causal relations based on which one may then explain – are necessarily *deductive*, as they always proceed (as Aristotle has put it) from the more general to the more particular (e.g., Piaget, 1950a: 265–341; 1963; 1967a: 766–772; 1970a: 47–49; 1970b: 233–234; 1971b: 37–44; Bunge, 1964, 1967: 3–65; , 1983: 3–16). Nevertheless, in spite of such differences, their perspectives on inter-level explanations largely coincide.⁴

⁴For an overview of this ‘revival’ of causality and causal explanations since the 1950s, with a special emphasis on the development of physics, biology, psychology, and the social sciences, see Bunge (1982). From the mid-1970s and 1980s, numerous other authors contributed to the rehabilitation of causal models of explanation, including Rom Harré, Roy Bhaskar, Michael Scriven, Peter Railton, Paul Humphreys, and Wesley Salmon. For a selective overview of this movement, though limited to the philosophical literature, see Salmon (1989).

Drawing on his work on developmental cognitive psychology and developmental epistemology, Piaget was one of the first scientists to explicitly recognize the need for an interlevel integrative model of explanation as an alternative to the reductionist models of explanation, equally supported by positivist and classical mechanistic philosophies. Piaget named his alternative model of explanation, ‘*reciprocal assimilation*’, ‘*reduction by interdependence*’, or ‘*hybridization*’ (1950b: 64–79; 1967b: 1151–1182 and 1249; 1970a: 46; 1970c: 469, 525).

According to Piaget, three types of dependence relations among theories addressing different levels of organization can be defended:

- (i) reduction from the ‘higher’ to the ‘lower’;
- (ii) irreducibility of the phenomenon of the ‘higher’ level; and
- (iii) reciprocal assimilation by partial reduction of the ‘higher’, but also by enrichment of the lower by the higher (1970c: 469).

In the latter case, “a more complex science can be integrated into a simpler one, but then it *enriches* the latter to *transform* it into a new system through the *interdependence* of the superior and the inferior” (Piaget, 1967b: 1182).

For Piaget, “even in physics attempts to reduce the complex to the simple, for example, electromagnetic to mechanical phenomena, lead to syntheses in which the more basic theory becomes *enriched* by the derived theory, and the resulting *reciprocal assimilation* reveals the existence of structures as distinct from additive complexes” (Piaget, 1971b: 45).⁵ In this line of thought, Piaget would go as far as writing that one can “be quite relaxed about the prospect that living phenomena will one day become reduced to physico-chemical ones; here, as in physics, reduction will not mean impoverishment but such *transformation* of the two terms connected as *benefits both*” (Piaget, 1971b: 45–46 – italics inserted). In other words, “if a physico-chemical explanation of life can be expected, our present physico-chemistry will gain *new properties* thereby, thus becoming more ‘general’ instead of being applied exclusively to more and more special fields” (1970c: 469 – italics inserted; also: Piaget, 1950b: 75–79).

A similar perspective was defended by Monique Lévy (1979) in the context of her analysis of the relationships between biology, chemistry, physical chemistry, and physics. Unlike reduction, taken as a purely “asymmetric relation”, *reduction by synthesis* “assigns specific theoretical roles to each discipline” (Lévy, 1980: 152–153). According to Lévy, as “a consequence of the specific role played by each of the disciplines partaking in the reduction, every ‘reduction by synthesis’ proceeds not by annexation of a domain in the frame of another, but by interaction, and by reciprocal enrichment” (Lévy, 1980: 153). For example, addressing the alleged reduction of chemistry to quantum mechanics, Lévy noted: “If we limit chemistry

⁵ Piaget presents as paradigmatic examples of these processes of reciprocal assimilation the simultaneous ‘geometrization’ of gravitation and ‘physicalization’ of the Riemannian space curves in Einstein’s general relativity theory, as well as the interactions between mechanics and electromagnetism, which, after a period of attempts at unilateral reduction, would then lead to the creation of wave mechanics (1967a: 768; 1970a: 46).

to what could be deduced from physics alone, whole areas of this science would disappear (kinetics, organic chemistry, biochemistry, non-equilibrium processes)” (1979: 348).^{6,7}

Piaget’s model of reciprocal assimilation (from 1950 onward), Lévy’s notion of reduction by synthesis (1979/1980), or Bunge’s views on integration (e.g., , 1983: 31–45 and 165–175) are precursor variants of what is presented today as ‘interlevel integrative explanations’ (e.g., Bechtel, 1986a; Craver, 2005; Brigandt, 2010; Craver & Darden, 2013: 161–185), ‘multiscalar’ or ‘multi-level contextual explanations’ (Bishop et al., 2022), or simply ‘interdependence’ and ‘hybridity’ (Cat, 2022).

In Craver and Darden’s (2013) classification of the different ways an integrative explanation can occur in a mechanistic explanation, ‘interlevel integration’ fulfils a special role. It consists in the integration of what different fields find at different levels of organization, “either by looking up to see how a phenomenon is integrated within higher-level mechanisms or by looking down to see how a phenomenon is integrated with lower-level mechanisms” (2013: 163). As Craver and Darden note, “many of the great achievements in the history of biology involve bridging different levels of mechanisms” (Craver & Darden, 2013: 167). Interlevel integration thus represents an alternative form of explanation to the classical micro- or lower-level reductionist models, and to their associated idea that fields studying lower-level phenomena are “always more fundamental in explanations” (Brigandt, 2010: 297).

Yet there is more to integration than simply putting forward integrative theories. Aside from integrating explanations, “integrating methods (inference and modeling methods as well as experimental methods) and integrating data” are required (Brigandt, 2013: 463). Philosophy thus needs to understand “how concepts, methods, and explanatory resources are in fact coordinated, such as in interdisciplinary research where the aim is to integrate different strands into an articulated whole” (Love & Lugar, 2013: 548). For example, evolutionary developmental biology, which is an attempt to promote a theoretical integration of evolutionary biology and developmental biology, “faces the significant challenge of integrating quite different methods and explanations, such as experimental and theoretical approaches, microevolutionary and macroevolutionary models, developmental and population genetic explanations” (Brigandt, 2010: 298).

⁶In 1977, Darden and Maull presented *interlevel* theories as a “subset” of interfield theories” (Maull, 1977: 160). Yet, they were more concerned with the relationships between the different methods, techniques, explanatory goals, and vocabularies of different ‘areas of research’ for solving some focal problems at different ‘levels of description’ than with theories of phenomena located at different “structural levels”, i.e., in terms of ‘part/whole’ orderings or relationships (Maull, 1977: 154; Darden & Maull, 1977: 44).

⁷According to Lévy, reduction by synthesis often results “in the construction of intermediary disciplines”. Therefore, “biology can now follow the example of chemistry in constructing those *intermediary* domains characteristic of reduction, without fearing mutilation; for a discipline is *enriched* when approached by the method of ‘reduction by synthesis’” (Lévy, 1980: 157 – italics inserted). For a recent account of Lévy’s notion of ‘reduction by synthesis’, see Bensaude-Vincent and Simon (2012: 164–168).

The set of problems and questions raised by the different forms of integration is so vast and complex that it cannot be discussed here. However, in the context of our analysis, the most important point issue to emphasize is that it is the objective existence of processes of systemic emergence and downward determination (causal or otherwise) that most strongly necessitates the use of interlevel integrative forms of explanation.

In order to explain emergent systems' properties and downward causal processes, we need to do more than make higher-level and lower-level descriptions compatible. We need an adequate articulation of upward and downward *explanations*. In other words, we need to explain (i) *why* lower-level systems behave the way they do as constitutive *relata* of some higher-level systems of relations, and (ii) *how* some system-level properties result from specifically organized systems of relations between their constitutive *relata* as lower-level parts. In sum, the critical issue is to explain how levels of organization *relate* to each other, *partially determining* each other's existence and identity.

12.4.2 *Inter-theoretical Relations*

We can also address this issue from the point of view of inter-theoretical relations. For example, under what conditions can a theory of a system *w*, of a kind *K*, be literally reduced to the theory about the lower-level individuals (*Ls*), of a kind *L*, that are or may be proper parts of *w*?

Can the theory of the intrinsic or *w*-independent properties, and respective causal and nomological relations, of *Ls*, fully explain all non-relational properties of system *w*? Only in this case, we could talk about a proper reduction of a *higher-level theory* to a *lower-level theory*. If some system properties of *w* are only completely determined or produced by (thereby only being fully explainable in terms of) a specific organization between some properties which *Ls* only acquired or manifest as parts of or *relata* within the specific system of relations which obtain in *w*, then in no meaningful way we can talk about a literal lower-level reduction. The fact that the complete explanation of *w*'s properties requires the *incorporation* of some *w*-dependent relational properties of *Ls* just means that the lower level is *not*, in itself and by itself, *enough* to account for all the higher-level properties of the system *w*.

Imagine that one could build a general theory (*G*) incorporating *all* properties that *Ls* can acquire or manifest in *every* possible relational context, including as proper parts of systems of the kind *K*. Then, if we abstract away from all exogenous relations that such systems may have with other systems (including in the context of further, higher-level systems of relations), we might say that all non-relational properties of systems of the kind *K* can be fully explained in terms of theory *G*. But what kind of explanation would that be? Could theory *G* be considered a literal lower-level theory, thereby enabling a proper lower-level reduction of the higher-level theory of systems of kind *K*? The answer can only be negative. Once a lower-level theory begins to incorporate all relational properties and behaviors that its systems acquire or manifest in *all* possible relational contexts, including as proper parts of higher-level systems,

it ceases to be a pure lower-level theory. That theory is already a *new* theory, changed and enriched by the integration of all higher-level relevant factors.

The theory-reduction model faces the same challenges as any mechanistic explanation that opts for a more ‘localized’ or ‘particularist’ approach.

Even if someone were to defend that all such system-dependent relational properties or causal powers were already possessed by the lower-level systems as *intrinsic* dispositions (as contemporary individualistic essentialists would argue), that would still not allow for proper lower-level reductive explanations. As I have argued elsewhere (Santos, 2020: 8695–8696; also: 2015a, 2021), even if the intrinsic dispositions thesis were right, one would still need to explain why some potentialities are actualized in certain relational settings but *not* in others, why some potentialities are actualized *instead* of others (including their direct opposites), and why some potentialities are not even *ever* actualized. To account for all this, we need relations and systems of relations as absolutely necessary *ontological* and *explanatory* factors. Indeed, without relations, including second-order and even higher-order ones, the alleged intrinsic potentialities of all lower-level entities would remain latent and inactive for all eternity. That is, they would never come into actual existence, thereby failing to have the causal effects that they actually have on the dynamics and structure of our universe.

Consider cell types. Cells differentiate depending on the properties of their developmental contexts. This means that if the same cells were put in a different tissue, they would differentiate differently (Vecchi, 2020b: 62–63). As Soto and Sonnenschein observed, “[a] single cell isolated from either one of these tissues (...) fails to originate the tissues that would result from their reciprocal interactions” (2011: 333). A particularly significant instance of RTE and DC is that when cancerous cells are transplanted or injected into healthy tissues, their behavior is ‘normalized’, reverting to a non-cancerous state (Soto & Sonnenschein, 2011: 338). How lower-level would then be a theory of cells integrating all properties that cells may acquire by virtue of being parts of or relata within *higher-level* organized systems of relations, such as *tissues* or *organisms*?

This problem, of course, is not new. According to Robert Causey, a lower-level reduction of a higher-level theory may be obtained if scientists “study the behavior of the components of structured wholes when they are *not* part of the whole (...) and then derive their behavior when part of the structured whole from this information *plus* specification of the boundary conditions prevailing when they are bound” (Bechtel & Hamilton, 2007: 398). Cliff Hooker and Patricia Churchland followed another, but similar strategy: “to incorporate into the lower-level theory everything that is learned about lower-level entities as they are bound into various structured wholes” (Bechtel & Hamilton, 2007: 398). In sum, “lower-level theories need to be enriched to account for what is learned at the higher-level” (Bechtel & Hamilton, 2007: 399).

As a matter of fact, neo-positivists were also well aware of the necessity of *changing* the lower-level theories by *enriching* them with all the necessary higher-level factors. For example, Nagel acknowledged that some systems (which he called ‘organic’ or ‘functional’) cannot be fully explained by and thus reduced to the laws relating the properties which their proper parts manifest *independently* of being parts of those systems. In such systems, parts “stand in relations of causal

interdependence”, that is, they “do not act, and do not possess characteristics, *independently* of one another” (Nagel, 1961: 395, 391). Therefore, “any laws which may hold for such parts when they are not members of a functional whole cannot be assumed to hold for them when they actually are members” (1961: 394). Any additive analysis of organic or functional wholes “must *include* special assumptions about the *actual organization* of parts in those wholes when it attempts to apply some fundamental theory to them”. In sum, the explanation of such systems “in terms of theories about their constituent parts cannot avoid *supplementing* these theories with statements about the special circumstances under which the constituents occur as elements in the systems” (1961: 395 – italics added).

Hempel also recognized the need to *supplement* the lower-level theories with information relative to specific higher-level systems of relations. In particular, Hempel argued that the complete explanation of some wholes could be provided only if in addition to the independent properties of their parts, we integrate all “relational information” concerning the “spatial or other relations”, including “structural relations”, among the parts (Hempel, 1965: 260–261). A complete explanation would then require a “description, in terms of relational concepts, of the way in which [parts] are connected with each other” in each different kind of whole (1965: 261).⁸

The problem with all these strategies is not so much the real possibility of building such general theories as the *epistemological meaning* of that possibility.

Consider, again, a general theory (*G*) that incorporates *all* properties that some entities (*Ls*) can acquire or manifest in *all* possible relational contexts, including as proper parts of systems of the kind *K*. If we took those systems as *isolated* and, for the sake of argument, we also ignored the historical processes that lead to the formation of their organizational structures, we could, of course, deliver a complete explanation of such system in terms of a *G* theory about *Ls*. The question is that *that* explanation would no longer represent a proper lower-level reduction. That theory would just be an example of an *interlevel integrative theory*, as the epistemological expression of the ontological reciprocal determination between lower and higher levels of organization.

The theory-reduction model attempted to assimilate that reciprocal determination by explicitly invoking the need of adding specific *boundary conditions* when reducing higher- to lower-level laws. The problem is that boundary conditions “are

⁸Hempel and Nagel were, of course, aware of the criticisms elaborated by their contemporary organismic views in biology, as well as by some holistic views in psychology (e.g., Gestalt theory) and in physics (e.g., the field theories). Bertalanffy, for example, criticized the “analytical, summative and machine theoretical viewpoints” of modern science, whose ultimate goal was “to explain phenomena by reducing them to an interplay of elementary units that could be investigated *independently* of each other” or “to resolve all natural phenomena into a play of elementary units, the characteristics of which remain *unaltered* whether they are investigated in *isolation* or in a *complex*”. In his view, the “organismic conceptions assert the necessity of investigating not only parts but also *relations of organisation* resulting from dynamic interaction and manifesting themselves by the *difference* in behavior of parts in isolation and in the whole organism” (1950: 134–135 – italics inserted).

not themselves derived from the lower-level laws". Additionally, "where do these boundary conditions come from?" (Bechtel & Hamilton, 2007: 399; see also: Bishop et al. 2022: 278–283).⁹

Some systems may be affected by same-level or even lower-level boundary conditions, but some boundary conditions are clearly the result of higher-level organizations (Noble et al., 2019). In the latter case, as Bechtel notes, "by just characterizing such information as specifying boundary conditions and not considering what that information is about, namely, the organization involved in constituting a *higher-level system* out of lower-level constituents, the theory-reduction account *camouflages* the contribution of higher-level inquiries" (Bechtel, 2007: 150, n. 6 – italics inserted).

Specific modes of organization are typically relegated to the status of boundary conditions by reductionists. Yet, as Bechtel observed,

insofar as the boundary conditions cohere into *stable structures* that are *heritable*, they acquire a significant status and must be accommodated in any general endeavor to describe the course of events. After they arise, some of these stable structures may be *perpetuated*", and "once it is recognized that these *organizational structures* are the result of an *historical process*, the significance of any attempt to give a reductionistic explanation is radically reduced. To complete the reduction, one must fill in the details of the boundary conditions as they have historically arisen, a task that *cannot* be completed with just the *laws of the basic theory* (Bechtel, 1986b: 97 – italics inserted).

Furthermore, according to Bishop (2019), in addition to the boundary conditions, we must also take into account the existence of *stability conditions*, which "don't function like boundary conditions", but rather constitute necessary conditions for the very "existence and persistence of appropriate states and observables and systems" (2019: 5.7). In fact, without specific stability conditions, there wouldn't even be nothing to which laws and boundary conditions could apply. And yet stability conditions are "never given by", nor they are "derivable from the underlying scale or domain alone" (Bishop, 2019: 3.2; also: Bishop et al. 2022: 27–36 and 275–278).¹⁰

⁹As Jean Ullmo has long ago observed, "[e]very law is relative to an isolated system, that is, it describes a specific type of interaction in which an object may be involved, abstraction made of all interactions that take place simultaneously, and which one assumes, or makes sure to be, negligible" (1958: 156). In this sense, all "repeatable relations" expressed by law-statements are "conditioned", and "it is [their] implicit conditions that ensure their validity" (1958: 54). The notion of scientific laws as exceptionless universal statements with an unlimited scope is a scientifically unfounded philosophical myth. Unfortunately, this caricatured interpretation of laws has often been used to minimize or even call into question the existence of laws in nature and the role they play in the construction of scientific knowledge.

¹⁰According to Bishop, Silberstein and Pexton, "Not only can stability conditions arise dynamically (and locally) constraining the behavior of basic components (e.g., convection), or be causal-mechanical constrains such as mechanical equilibrium, but stability conditions also include global or systemic constrains such as topological constrains, dimensional constrains, network or graphical constrains, and order parameters, among others. One can point to both dynamical and adynamical, causal and acausal, as well as local and global stability conditions that are difference makers. For example, global adynamical constrains might include conservation laws, free energy principles, least action principles, symmetries, and some types of symmetry breaking" (2022: 28).

All of this shows the reason why the causal closure or completeness principle of any level of organization should be discarded as simply wrong (Bishop et al., 2022: 288–303).

Another related issue (already mentioned) comes with the fact that, very often, the accounts that present themselves as constituting lower-level explanations are grounding their explanations in lower-level individuals taken *as already determined* parts of or relata within specific higher-level systems. Sarkar (2015) points to this problem in Nagel’s model of inter-theoretical reduction. Reductions often involve approximations and, in particular, approximations on which the derivation of higher-level from lower-level theories depend. Still, there are approximations that “are justified by the reducing theory”, while others “are tailored to fit the reduction and implicitly rely on the reduced theory”. As Sarkar notes, “the serious question [is] whether a reduction *only* invokes as *explanans* factors that are indubitably from the reducing theory” (2015: 50). In sum, the problem is that of *presupposing* higher-level determining factors when we elaborate our lower-level theories.

As Bishop, Silberstein and Pexton note, many of the “alleged” inter-theoretic reductions require to “implicitly import some of the wider contextual features at the higher-level without acknowledging them” (2022: 69).

This problem is debated in most if not all sciences. Consider a prototypical example of a holist explanation in the social sciences: “the rise in unemployment led to a higher crime rate”. Now consider the alternative individualist explanation: “as a result of individuals *a*, *b*, *c*, etc. losing their job and feeling very frustrated about having little money and no job opportunities, the crime rate went up” (Zahle & Kincaid, 2019: 659). The relevant question in this context is *how micro- or low-level* is an explanation of a social phenomenon that adduces in its *explanans* facts such as ‘having little money’ and ‘having no job opportunities’. These two properties are clearly not psychological. Both are extrinsic relational properties that individuals acquire solely by virtue of living in a specifically organized structure of socio-economic relations. Indeed, some properties combine properties pertaining to different levels of organization. Think of a complex property such as ‘being afraid of losing her job’. What would constitute a real explanation of the instantiation of that complex property? The first property is clearly psychological, referring to a particular mental state, while the second is a clear socio-economic extrinsic relational property. As Auyang pointed out, “[a]lmost all explanations in terms of concrete individuals involve social predicates. Thus, social concepts have not been eliminated as demanded by methodological individualism; *they have only been swept under the rug*” (1999: 357, n.1 – italics inserted). As Kincaid has noted, the problem is that of “presupposing the reduced theory in the reducing explanations” (2012: 148), that being the reason why “many so called individualist explanations are really individualist only in name” (2012: 149).¹¹

¹¹ Margaret Archer (1995)’s morphogenetic social theory is an attempt to overcome both individualistic and holistic approaches in contemporary sociological thinking.

Nevertheless, from the vantage point of a relational view, there is no reason to see a conflict between higher-level explanations (e.g., sociological) in terms of some wholes (e.g., systems of social relations) and lower-level explanations (e.g., psychological) in terms of their proper parts (e.g., human beings), for they “complement each other in revealing the dual aspect, individual and inter-individual, of all behaviour patterns in human society” (Piaget, 1995: 41).

That complementarity is clearly exemplified in the set of all *system-dependent relational properties* that entities acquire and manifest as *relata* of specifically organized systems of interdependent transformative relations. Just as each human being is an already *socialized* individual when living in a specifically organized system of social relations, each atom is an already *molecularized* atom when part of a specifically organized molecular system of interdependent relations, and each molecule (such as DNA) is an already *cellularized* molecule when part of a specifically organized cellular system of interdependent relations. This is how we should conceptualize individual entities as *relata* within integrated relational systems. For example, a human being is not only a bio-neuro-psychological system but a *socio-economically* and *culturally shaped* bio-neuro-psychological being (e.g., Archer, 1995; Bishop et al., 2022: 223–226 and Lawson, 2019).¹²

A higher-level organization cannot be just anything, that is, regardless of the entities composing it (one cannot build living cells with crystals). That is why there is always a partial micro-determination of the higher levels. But the lower-level properties and laws are not enough to completely determine, and thus to fully explain, not just all higher-level systems but also the behaviors of the lower-level entities as parts of such systems. In sum,

The arrow of determination and explanation is not exclusively bottom-up but multi-scale and multidirectional, since any causal process is bounded by relational constraints which can be top-down, bottom-up, or side to side (as it were). There are no discrete causally closed or absolutely autonomous scales or domains of reality. Rather, there is a relation of mutual integration, interdependence, and reciprocal conditioning (Bishop et al., 2022: 283).

This is the reason why in many, if not in all cases, “the higher-level theories (for instance, cell physiology) and the lower-level theories (for instance, biochemistry) are ontologically and epistemologically inter-dependent on matters of informational content and evidential relevance” (Cat, 2022). In other words, scientific explanations often require “a genuine ‘*hybridization*’, with fruitful re-combinations”, between different disciplines or domains of research, where “the link between a ‘higher’ (in the sense of ‘more complex’) and a ‘lower’ field results neither in a reduction of the first to the second nor in greater heterogeneity of the first, but in *mutual assimilation* such that the second explains the first, but does so by *enriching* itself with *properties* not previously perceived, which afford the *necessary link*” (Piaget, 1970c: 525 – italics inserted).

¹²In this regard, it is worth mentioning that the causal impact of socioeconomic factors on brain structure and functioning, as well as cognitive and emotional development, is a growing research topic in current *social neuroscience* (e.g., Troller-Renfree et al., 2022; Thomas & Coecke, 2023).

This process of enrichment/reciprocal assimilation is paradigmatic in cases of real interlevel integration. For example, according to biophysicist Henri Atlan, it is not true that ‘life’ has been literally reduced to physical chemistry. What happened is that physical chemistry was changed and “extended”, thereby allowing the creation of a “biophysics of organized systems” (Atlan, 1979: 23–24). Similarly, biochemistry is not just ‘applied chemistry’, since it already constitutes “an *extension* relative to mineral and organic chemistry” (Atlan, 1979: 24, n.1; see also Bechtel, 1986b). And the same could be said about the relationships between quantum mechanics and chemistry, since the laws of the former do not fully determine quantum chemistry (Bishop, 2019, sections 4.13–4.18).

The process of *enriching* lower-level theories, as well as the construction of intermediate theories between lower- and higher-level theories, is just the *epistemological* replication of the ontological process of *transformation* that lower-level entities undergo in terms of the *relational* properties they acquire and manifest as relata within higher-level structured systems of relations.

12.4.3 Some Implications for a Neo-mechanistic Model of Explanation

This process of enrichment/reciprocal assimilation is also evident in any neo-mechanistic model of explanation. At least in the case of integrated systems, *reduction* only refers to the necessary, but in itself insufficient, *analytical* methodological step of the explanation process concerning the operations of decomposition and localization. First, because these operations must be followed by two other methodological steps, namely, the *synthetic* operations of ‘recomposing’ and ‘situating’ a mechanism as a whole (see Sect. 12.2). Second, because these last two operations often show that the previous decompositions and localizations must be revised and corrected (Bechtel & Richardson, 2010: xxxvii–xl, *et passim*). Therefore, the explanatory tasks of recomposing and situating determine the very operations of decomposition and localization by showing the ways in which these may (and may not) be carried out (Bechtel, 2002; Bechtel & Abrahamsen, 2010; Bishop et al., 2022: 235–239).

Two conclusions can be drawn from this. First, the methodological steps of a neo-mechanistic model of explanation do not progress themselves in a linear, sequential fashion, but rather constitute a cyclic process with feedback consequences. Second, the explanatory task of ‘situating’ should be applied not only to a system or mechanism as a whole, but *also* to its *parts* since parts are also dependent both on intra- and extra-systemic relations. As it was argued, the properties and laws of any class of entities, taken as independent beings or isolated systems, only provide the necessary but not the sufficient conditions, not just to ontologically determine any higher-level organization but also to determine their own existence and identity.

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